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Natural Language Processing and Computational Linguistics

Course Code: CSI4001

NAME: DANDUSHAIK MAHEWISH

REG NO:23MIC0062

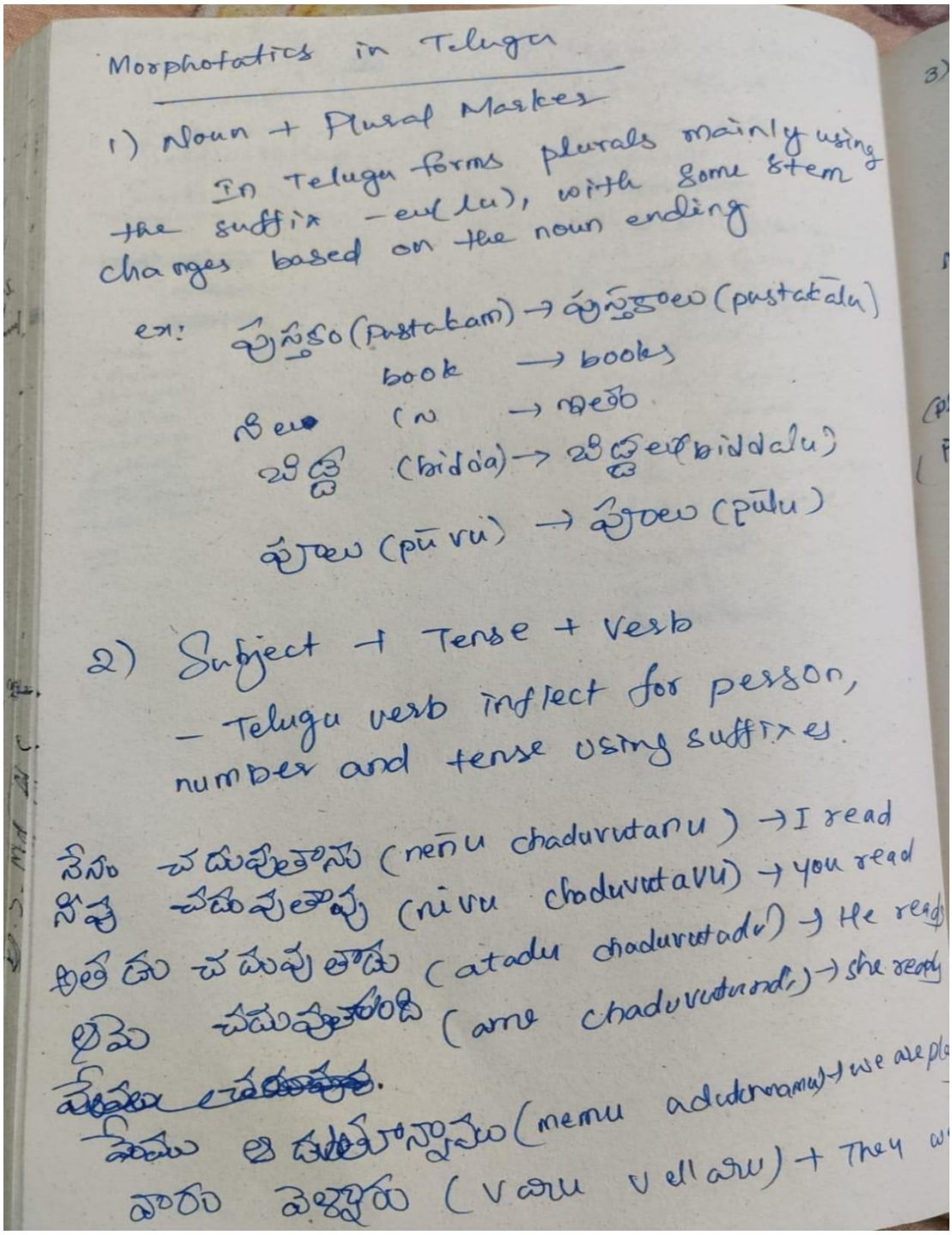
NLP PORTFOLIO

Under the guidance of Prof. Sharmila Banu

1) I was appreciated for writing an appropriate conclusion for Question 4 in CAT 1, demonstrating my understanding of the topic. (1 star)

Answer: I successfully presented a clear and relevant conclusion covering the key points of the question.

2) Takeaway assignment – Morphotactic (3 stars)

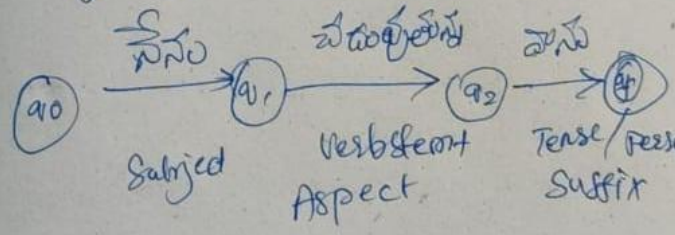


3) Adjective agreement with noun:
- Adjective agree with the gender and numbers of the noun

Gender	Noun	Adjective	Example
Masculine (Sin)	వాడు	పెద్ద	పెద్ద వాడు (big boy)
Femine (Sg)	వాళ్ళ	పెద్ద	పెద్ద వాళ్ళ (big girl)
Neuter (Sg)	ఇల్లు	పెద్ద	పెద్ద ఇల్లు (big house)
(Pl) Masculine	వాళ్ళు	పెద్ద	పెద్ద వాళ్ళు (big boys)
(Pl) Feminine	వాళ్ళలు	పెద్ద	పెద్ద వాళ్ళలు (big girls)

4) Morphological Structure (Finite-State Representation)

eg: నేను చదువుతూ ఉన్నాను (I am reading)





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Natural Language Processing and Computational Linguistics

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NLP PROJECT WRITE-UP

NAME: Dandushaik Mahewish

Reg. No.: 23MIC0062

LEXIMIND: AI-Powered Legal Document Simplification and Contract Analysis for Telugu and English

TEAM MEMBERS: Dandushaik Mahewish - Reg. No.: 23MIC0062

B. Pavani Yadav-Reg no:23MID0333

1. TITLE

LexiMind is an NLP-based web application designed to simplify complex legal documents and make contractual information accessible to ordinary users. It supports Telugu and English documents and provides same-language simplification, clause classification, key information extraction, potential-risk identification and document-based question answering.

2. MOTIVATION

Legal agreements such as rental agreements, sale deeds, lease agreements, affidavits and non-disclosure agreements often contain complex terminology, lengthy sentences and archaic expressions. The motivation behind LexiMind is to make legal information easier to understand for people without a legal background.

- Legal accessibility: Help ordinary individuals understand contractual rights, duties and obligations without requiring advanced legal knowledge.
- Regional-language support: Enable Telugu-speaking users to understand legal documents in simple Telugu instead of depending on English translations.
- Time efficiency: Reduce the time required to manually read lengthy legal agreements.
- Risk awareness: Identify potentially unfavorable conditions, penalties, termination clauses and financial obligations.
- Reliable information: Extract names, dates and monetary values without inventing information absent from the original document.

3. EXISTING GAP / EVENT THAT MOTIVATED THE PROJECT

Existing document-processing and general-purpose AI tools have limitations when handling regional-language legal agreements.

- Language barrier: Many tools translate Telugu documents into English instead of simplifying them directly into conversational Telugu.
- Complex legal terminology: Generic summarization tools may overlook important contractual conditions and legal obligations.
- Scanned documents: Traditional legal documents are often available as scanned PDFs or photographs, requiring OCR before processing.
- Lack of structured analysis: General-purpose chatbots do not necessarily provide dedicated clause classification, party-wise obligations, financial extraction and risk highlighting.
- Factual reliability: AI-generated responses may introduce information that is not explicitly mentioned in the original agreement.

Proposed Solution

LexiMind addresses these gaps through an integrated NLP workflow combining OCR, language detection, text simplification, clause analysis, entity extraction, heuristic risk detection and document-grounded question answering. Its primary functionality preserves the original language while simplifying the content.

4. ARCHITECTURE

LexiMind uses a modular, layered architecture. The Streamlit interface coordinates the complete workflow, while dedicated Python modules perform document extraction, language processing, text simplification, clause analysis, risk detection, retrieval-based question answering and report generation. The system is designed to process one uploaded document at a time and supports Telugu, English and mixed-language content.

4.1 End-to-End System Architecture

The system receives a legal document from the user, extracts text from digital or scanned files, preprocesses the content and passes it through the NLP core. The NLP core performs language detection, simplification, clause classification, entity extraction and heuristic risk assessment. In parallel, document chunks are indexed for retrieval-based question answering. Results are presented through the Streamlit interface and can be exported in multiple formats.

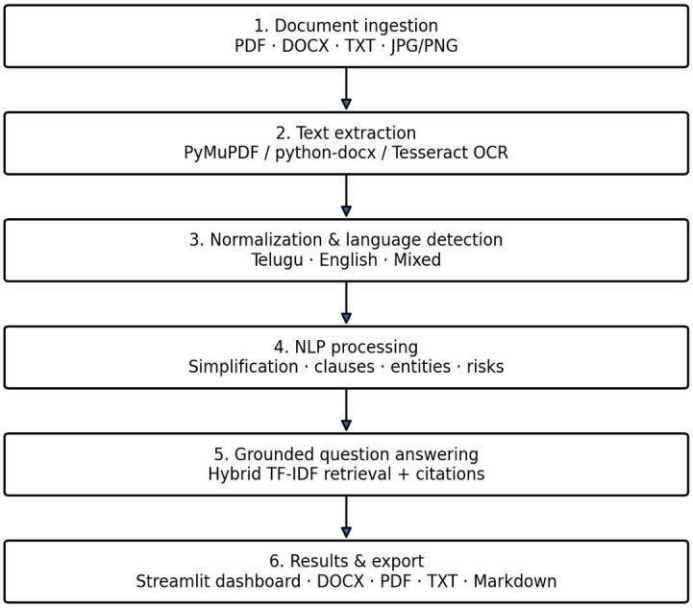


Figure 4.1. LexiMind end-to-end processing architecture.

4.2 Architectural Layers

The architecture is organized into three implementation layers and five functional subsystems. The implementation layers describe how the application is deployed, while the functional subsystems describe the processing responsibilities.

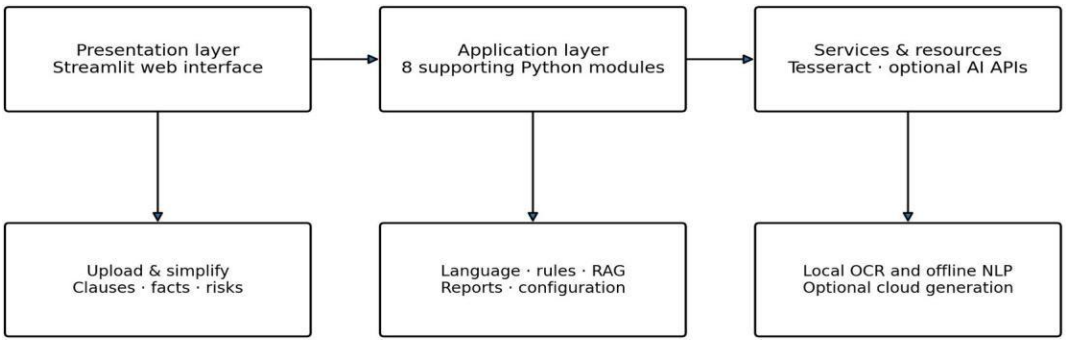
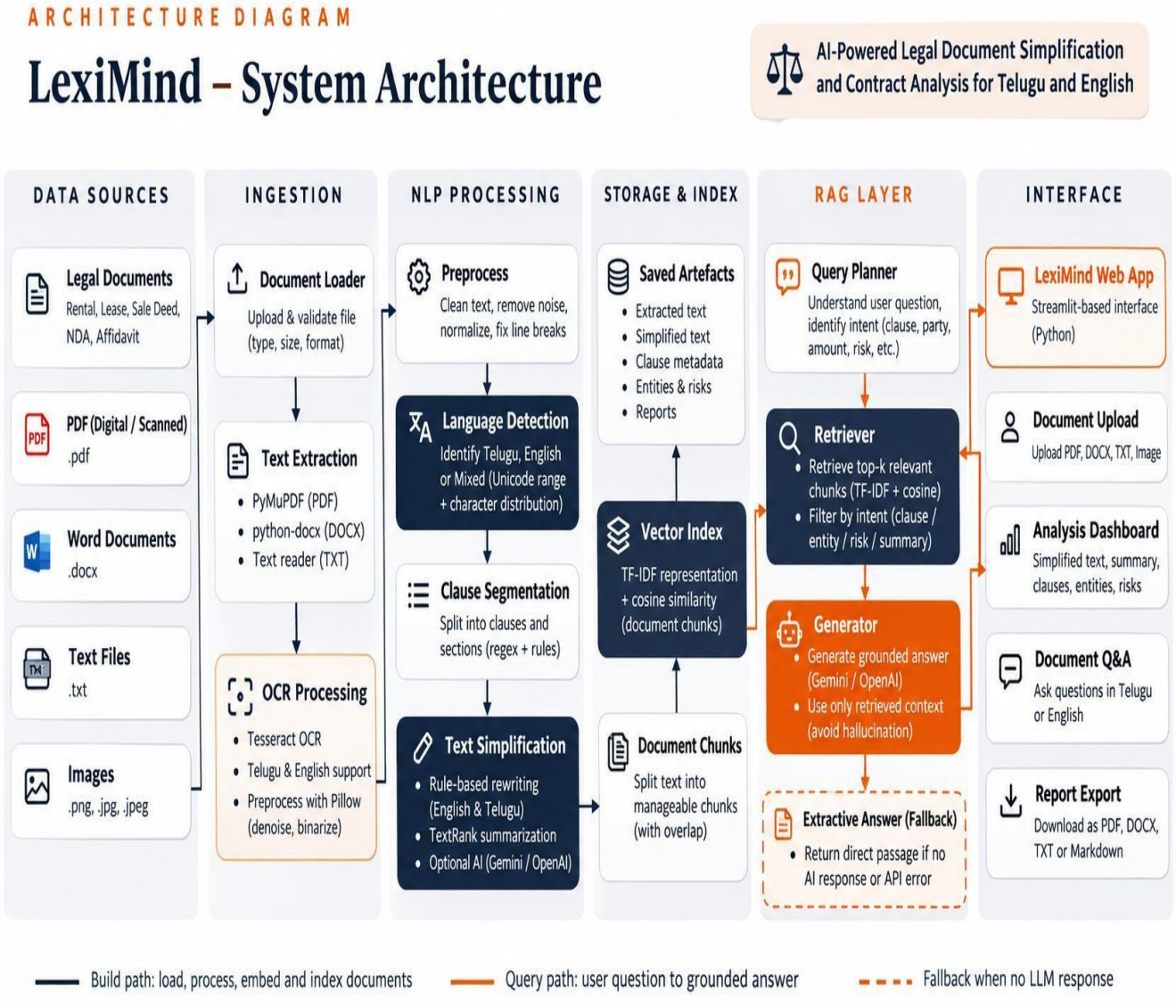


Figure 4.2. Presentation, application and service layers.

4.3 LexiMind System Architecture Diagram,



4.3 Functional Subsystems

A. Document Ingestion and OCR Subsystem

Accepts PDF, DOCX, TXT and image files. Digital PDF text is extracted using PyMuPDF, Word content is read using python-docx, and text files are decoded using supported encodings. Scanned PDFs and images are preprocessed with Pillow and passed to Tesseract OCR using Telugu and English language data.

B. Language Processing and Simplification Subsystem

Normalizes extracted text, detects Telugu, English or mixed-language content, and simplifies legal terminology while preserving the original language. The offline engine applies curated English and Telugu rewriting rules and generates extractive summaries using TextRank. Optional generative-AI services can improve natural-language rewriting when configured.

C. Contract Intelligence Subsystem

Identifies clause boundaries and classifies contractual sections. It extracts key facts such as party names and roles, dates, monetary amounts, agreement periods, notice periods and property identifiers. A rule-based risk analyzer highlights potentially unfavorable conditions and provides the source passage, reason and plain-language explanation.

D. Retrieval and Document Question-Answering Subsystem

Splits the uploaded document into clause-aware chunks and represents the text using TF-IDF features. Word and character n-grams, query expansion and cosine similarity are used to retrieve relevant passages. Answers are grounded in the retrieved text; when the document does not contain the requested information, the system is designed to return an explicit not-found response.

E. Presentation and Report Generation Subsystem

The Streamlit application displays simplified text, summaries, classified clauses, extracted facts, risk flags and document-grounded answers. The report generator prepares downloadable DOCX, PDF, TXT and Markdown files.

4.4 Application Modules and Responsibilities

4.5 Services, Data Handling and External Integration

- Tesseract OCR: Local OCR engine configured with Telugu and English language models for scanned documents and images.
- Generative AI services: Optional Gemini and OpenAI API integration for enhanced language rewriting; API use depends on environment configuration and network availability.
- Offline processing: Rule-based simplification and TextRank summarization remain available without cloud credentials.
- Document handling: Uploaded content is processed in memory and is not permanently stored by default, as described in the project design.
- Configuration: API credentials and application settings are supplied through environment variables rather than hard-coded in source files.

4.6 Seven-Stage Processing Pipeline

The complete document workflow is divided into seven stages, beginning with document ingestion and ending with grounded question answering and report export.

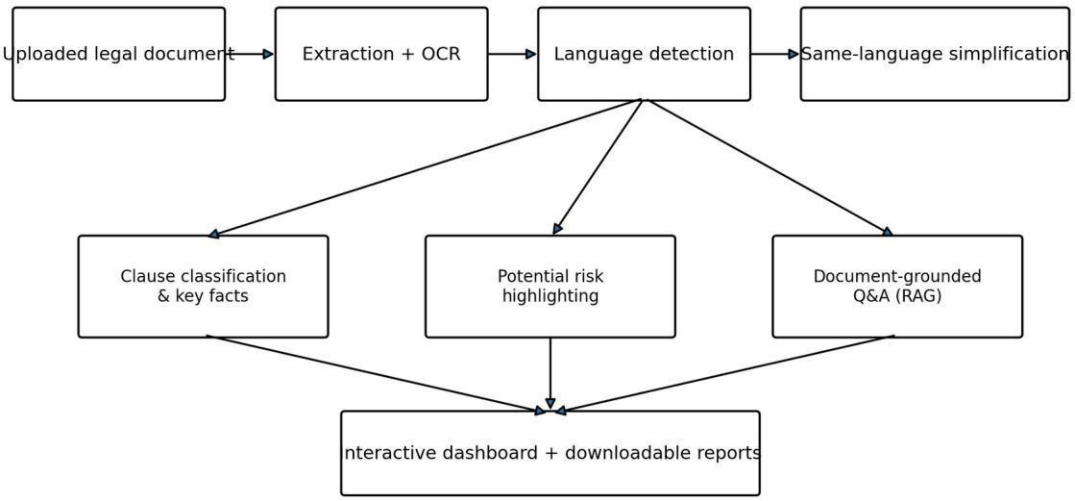


Figure 4.3. Seven-stage processing workflow from document upload to downloadable output.

Stage 1 - Ingest: Accept PDF, DOCX, TXT and image inputs, validate the uploaded file, extract digital text and apply OCR to scanned documents.

Stage 2 - Normalize: Repair broken line wraps, remove unnecessary OCR noise, normalize sentence formatting and prepare the extracted text for analysis.

Stage 3 - Detect Language: Inspect character distribution using the Telugu Unicode block (U+0C00-U+0C7F) and English alphabetic characters to identify Telugu, English or mixed-language content.

Stage 4 - Segment: Identify headings, numbered clauses, contractual sections and signature blocks so that the document can be analyzed in meaningful units.

Stage 5 - Simplify: Rewrite English legalese and archaic Telugu into plain language, preserve important facts and generate an extractive summary using TextRank. Optional AI rewriting may be used when available.

Stage 6 - Extract and Assess: Extract parties, monetary amounts, dates, agreement periods, duties and property details, then apply heuristic rules to highlight potentially unfavorable clauses.

Stage 7 - Answer and Export: Retrieve relevant document passages using TF-IDF and cosine similarity, answer user questions with document-grounded information and export the results as DOCX, PDF, TXT or Markdown.

5. COMPLETE TECHNOLOGY STACK

Component	Technology	Purpose
Programming Language	Python 3.11	Backend implementation
Frontend / UI	Streamlit	Interactive web application
PDF Processing	PyMuPDF	Extract text from PDFs
Word Processing	python-docx	Read and generate DOCX files
Image Processing	Pillow	Image preprocessing
OCR	Tesseract / pytesseract	Telugu and English text recognition
NLP / Machine Learning	Scikit-learn	TF-IDF and cosine similarity
Numerical Processing	NumPy	Numerical operations

Data Processing	Pandas	Structured data handling
Text Processing	Regular Expressions	Pattern extraction and normalization
Summarization	TextRank	Extractive summarization
Generative AI	Google Gemini API	Optional AI-based simplification
Alternative AI Provider	OpenAI API	Optional AI-based processing
PDF Generation	ReportLab	PDF report generation
Telugu Text Rendering	uharfbuzz	Unicode text shaping
Configuration	python-dotenv	Environment variables and API configuration
Testing	Pytest	Unit and integration testing

Main Python Modules

- app.py – Main Streamlit application.
- config.py – Configuration and environment settings.
- document_processor.py – Document extraction and OCR.
- language_processor.py – Language detection and AI integration.
- text_simplifier.py – Offline NLP simplification and summarization.
- clause_analyzer.py – Clause classification and entity extraction.
- risk_analyzer.py – Potential-risk identification.
- rag_engine.py – Retrieval and document-based question answering.
- report_generator.py – Report generation and export.

6. DATASETS USED

LexiMind uses sample legal documents for testing and reference legal NLP datasets for designing its classification, extraction and retrieval approaches.

A. Sample Evaluation Documents

The project documentation describes nine evaluation documents: four English legal documents, three Telugu legal documents, one mixed Telugu-English document and one scanned Sada Bainama document used for OCR testing. The examples cover rental agreements, lease agreements, sale deeds, NDAs and affidavits.

B. Reference Datasets and Resources

Dataset / Resource	Description	Usage in LexiMind
CUAD	510 commercial contracts with 41 clause categories	Clause taxonomy and risk-rule design
ContractNLI	Contract-based natural language inference dataset	Document-grounded Q&A and missing-information handling
LegalBench	162 legal reasoning tasks	Legal reasoning and evaluation reference
ILDC	Indian Legal Documents Corpus	Indian legal terminology and document conventions

IndicNLP Resources	Indian-language NLP resources	Telugu language and Unicode processing
Tesseract Telugu Data	Telugu OCR trained data	Recognition of Telugu characters in scanned documents

C. Custom Resources

- 174 text-rewriting rules: 114 English and 60 Telugu.
- 57 legal glossary terms: 32 English and 25 Telugu.

Dataset Usage Clarification

The reference datasets informed design and rule development. The project does not claim to have trained or fine-tuned a new machine learning model using these datasets. Evaluation documents are used for testing the implemented application.

7. LIST OF PROMPTS USED FOR THE CONSTRUCTION OF THE WEBSITE

LexiMind is implemented primarily using Python and Streamlit. Therefore, the website interface itself is not generated by a prompt-based website builder. However, the application includes an optional generative-AI prompt for document simplification.

A. Main Legal Document Simplification Prompt

> *"You are a legal document simplification assistant. Rewrite the legal document below so an ordinary person can understand it."*

The prompt includes the following instructions:

1. Output the content in the requested language.
2. Simplify legal terminology and complex sentences.
3. Preserve names, dates, amounts, numbers and property identifiers exactly.
4. Use only the information available in the uploaded document.
5. Clearly mark missing or unreadable information as “Not found”.
6. Explain the parties, financial details, dates, obligations and potentially risky clauses.
7. Avoid adding generic legal advice that is not supported by the document.

B. Language-Specific Instructions

- Telugu: Rewrite the document in simple, modern spoken Telugu without unnecessarily translating it into English.
- English: Rewrite the document in plain, everyday English.
- Translation: When explicitly requested, translate the document's meaning into the selected language while simplifying it.

C. Explanation-Level Instructions

- Simple Mode: Provide a short explanation that an ordinary person can understand quickly.
- Detailed Mode: Explain each clause, the affected parties, conditions and exceptions.

D. Retrieval-Based Question Answering Instructions

- Retrieve relevant passages from the uploaded document.

- Use the retrieved information to answer the user's question.
- Maintain document-based factual accuracy.
- Do not invent missing contractual details.
- Return “Information not found” when the document does not support an answer.

Note: These are the application's documented runtime AI prompt instructions and prompt structure, not a claim that these exact prompts were used to generate the website during development.

8. NLP PIPELINE

LexiMind follows a seven-stage NLP processing pipeline.

Stage 1: Document Ingestion

- Accept PDF, DOCX, TXT and image inputs.
- Validate the uploaded file.
- Extract digital document text.
- Apply OCR to scanned documents and images.

Stage 2: Text Normalization

- Remove unnecessary OCR noise.
- Repair broken lines and sentence formatting.
- Normalize extracted content.
- Prepare text for further NLP operations.

Stage 3: Language Detection

- Identify Telugu characters using Unicode range U+0C00 to U+0C7F.
- Identify English alphabetic characters.
- Calculate character distribution.
- Classify the document as Telugu, English or mixed-language.

Stage 4: Clause Segmentation

- Identify headings and numbered clauses.
- Separate individual contractual sections.
- Recognize signature blocks.
- Classify clauses into documented legal categories.

Stage 5: Text Simplification and Summarization

- Apply English legalese rewriting rules.
- Convert archaic Telugu into modern conversational Telugu.
- Generate summaries using TextRank.
- Extract duties, rights and prohibitions.
- Calculate readability statistics.

Stage 6: Entity Extraction and Risk Analysis

- Extract parties and their roles.
- Identify monetary amounts, dates and time periods.

- Extract property identifiers.
- Detect potentially unfavorable clauses.
- Generate explanations and review suggestions.

Stage 7: Retrieval, Question Answering and Export

- Divide the document into relevant chunks.
- Apply TF-IDF vectorization.
- Calculate cosine similarity.
- Retrieve relevant passages.
- Generate grounded answers.
- Export reports in DOCX, PDF, TXT and Markdown formats.

9. ELABORATE ON USERS, INPUTS (TYPES, FORMATS), OUTPUT

A. Target Users

- Tenants and landlords who need to understand rental agreements.
- Property buyers and sellers who need to understand sale deeds.
- Small business owners reviewing contractual agreements.
- Individuals handling affidavits and other legal documents.
- Telugu-speaking users who prefer legal explanations in their regional language.
- Individuals without formal legal knowledge who need accessible explanations.

B. Inputs

Input Type	Supported Format	Purpose
Digital Documents	PDF	Extract contractual information
Word Documents	DOCX	Process editable legal agreements
Plain Text	TXT	Process text-based agreements
Scanned Documents	Scanned PDF	Extract text using OCR
Images	PNG, JPG, JPEG	Recognize text using OCR
User Queries	Natural-language questions	Ask questions about uploaded documents
Language Selection	Telugu / English	Select simplification or optional translation language
Simplification Level	Simple / Detailed	Control explanation depth

C. Processing

The uploaded document undergoes extraction, normalization, language detection, clause segmentation, simplification, entity extraction, risk identification and document-based retrieval. Users can inspect and correct extracted text before further analysis.

D. Outputs

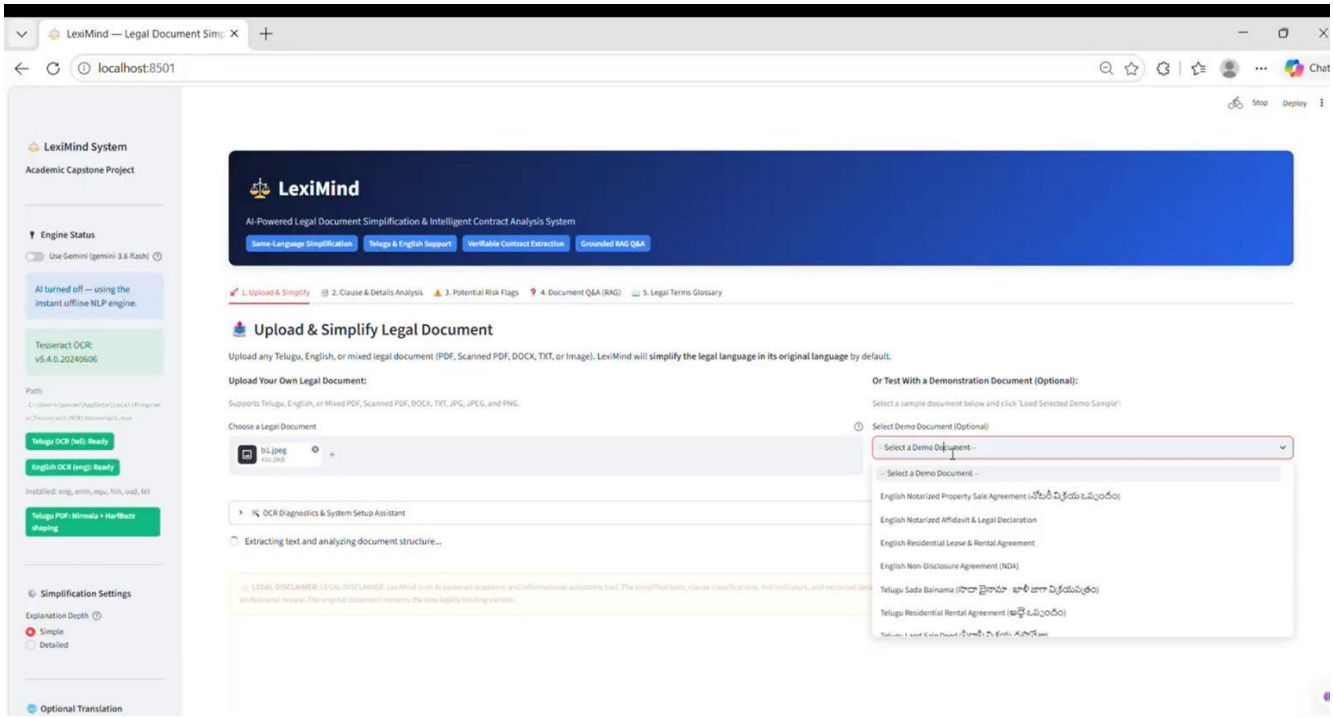
- Simplified Document: Original legal wording converted into understandable Telugu or English.

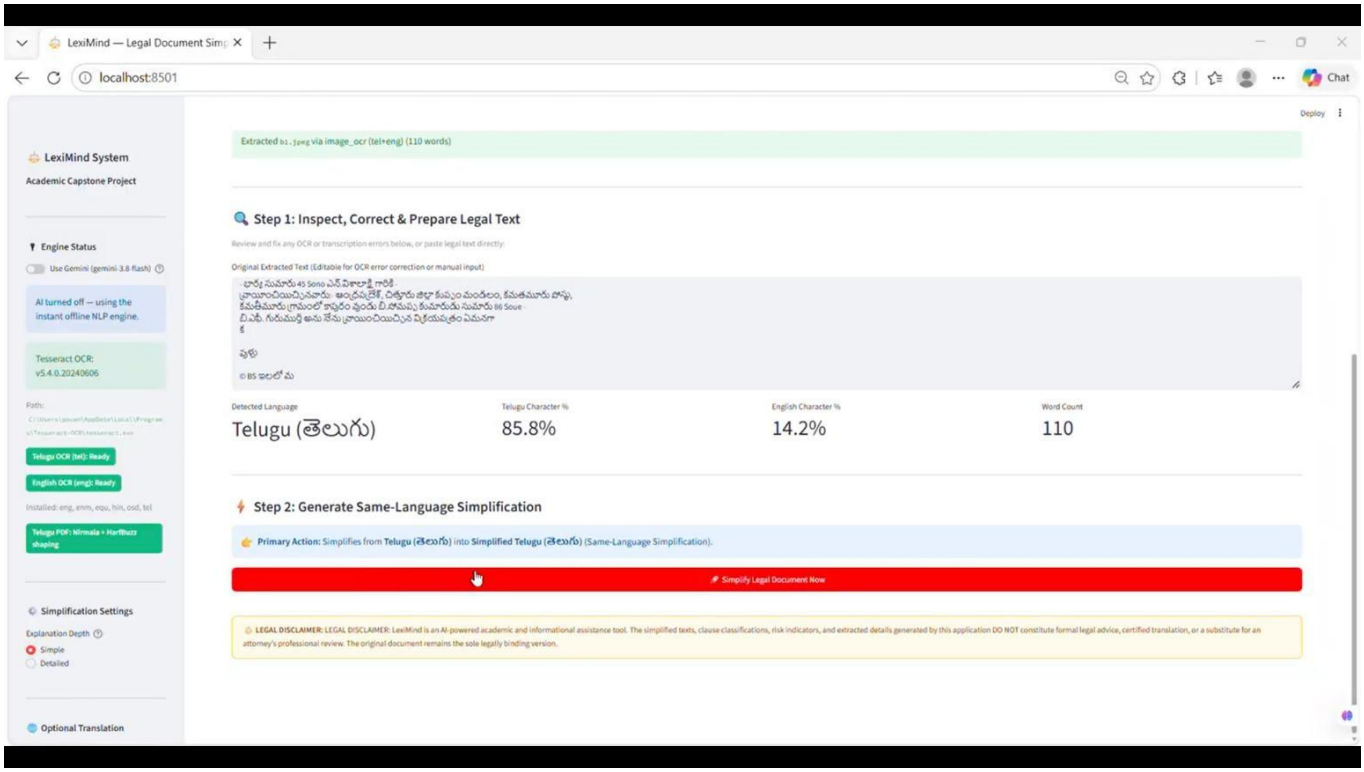
- Document Summary: Important information presented in concise points.
- Clause Classification: Contractual clauses grouped into relevant legal categories.
- Entity Extraction: Parties, dates, financial values, agreement periods and property details.
- Risk Analysis: Potentially unfavorable clauses highlighted with explanations and suggestions for review.
- Document Q&A: Answers to user questions based on relevant passages from the uploaded document.
- Legal Glossary: Difficult legal terminology explained in simple language.
- Downloadable Reports: DOCX, PDF, TXT and Markdown formats.

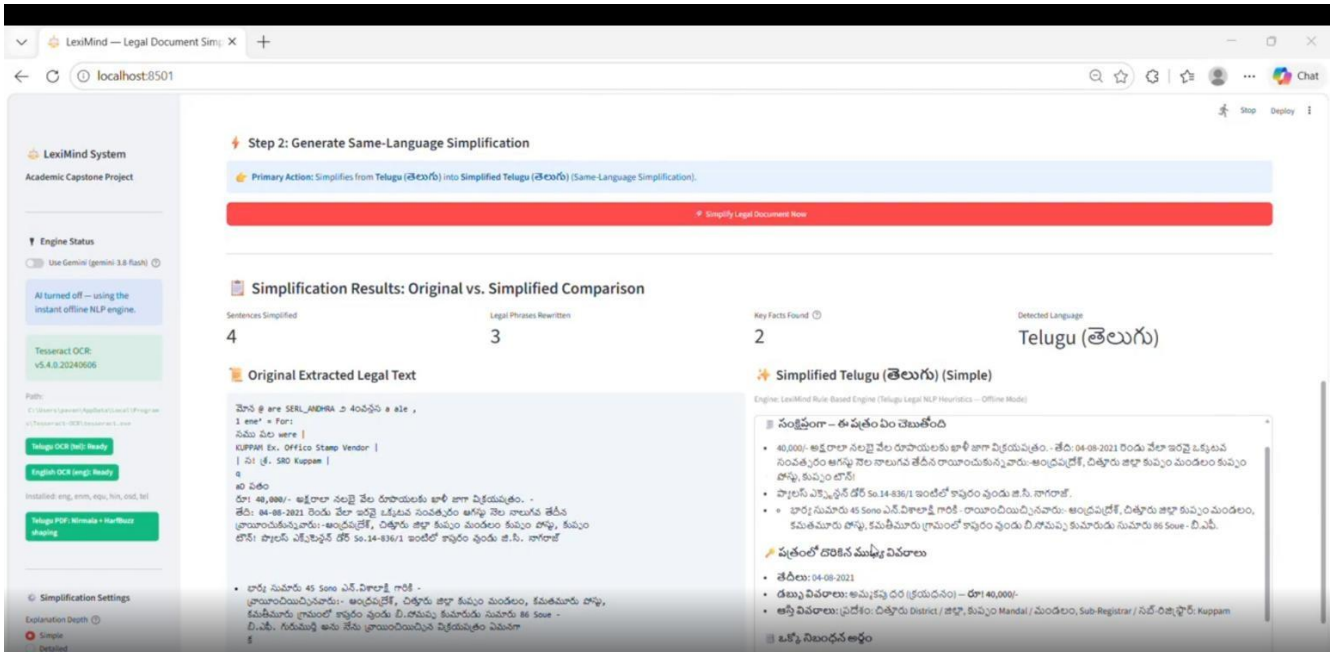
E. Expected Outcome

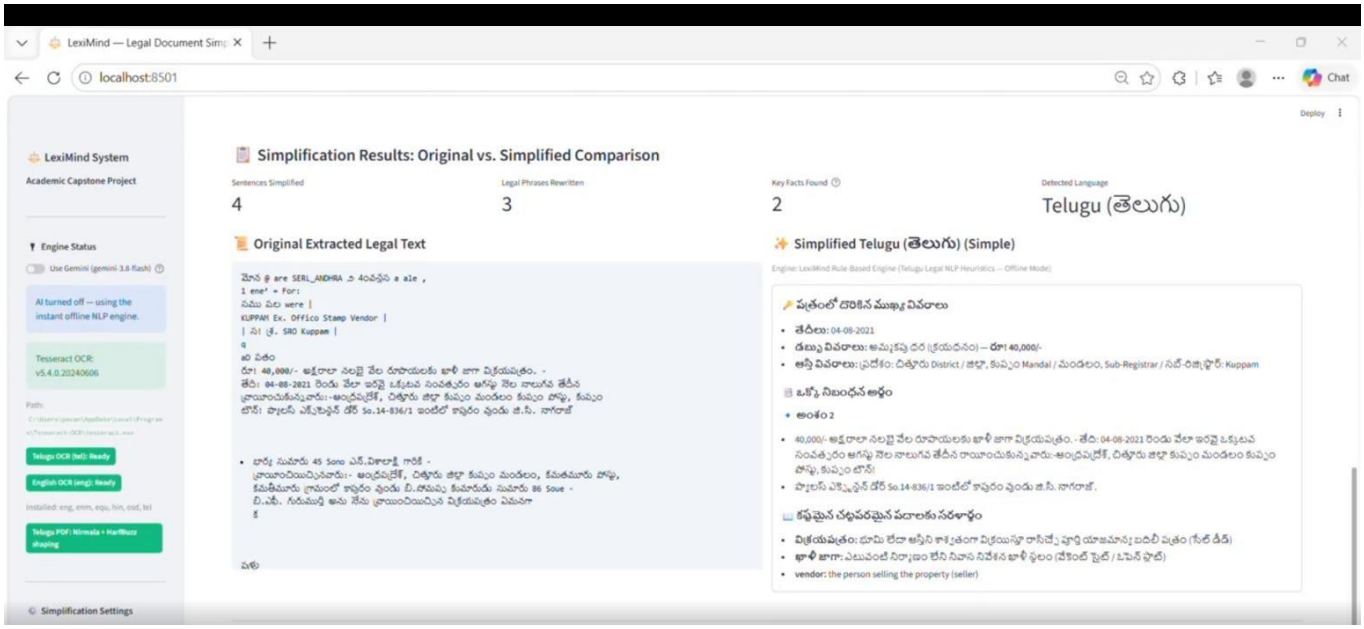
LexiMind provides an integrated platform that helps users understand legal agreements, identify important contractual information and recognize conditions that may require further attention. It is an informational assistance tool and does not replace professional legal advice.

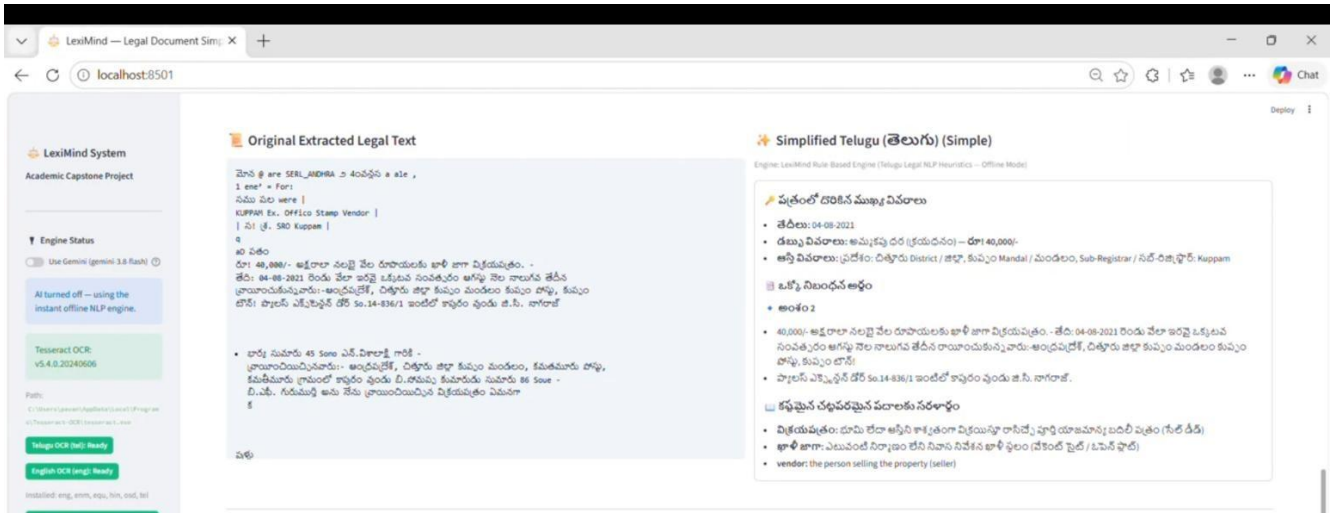
Input and output screenshots

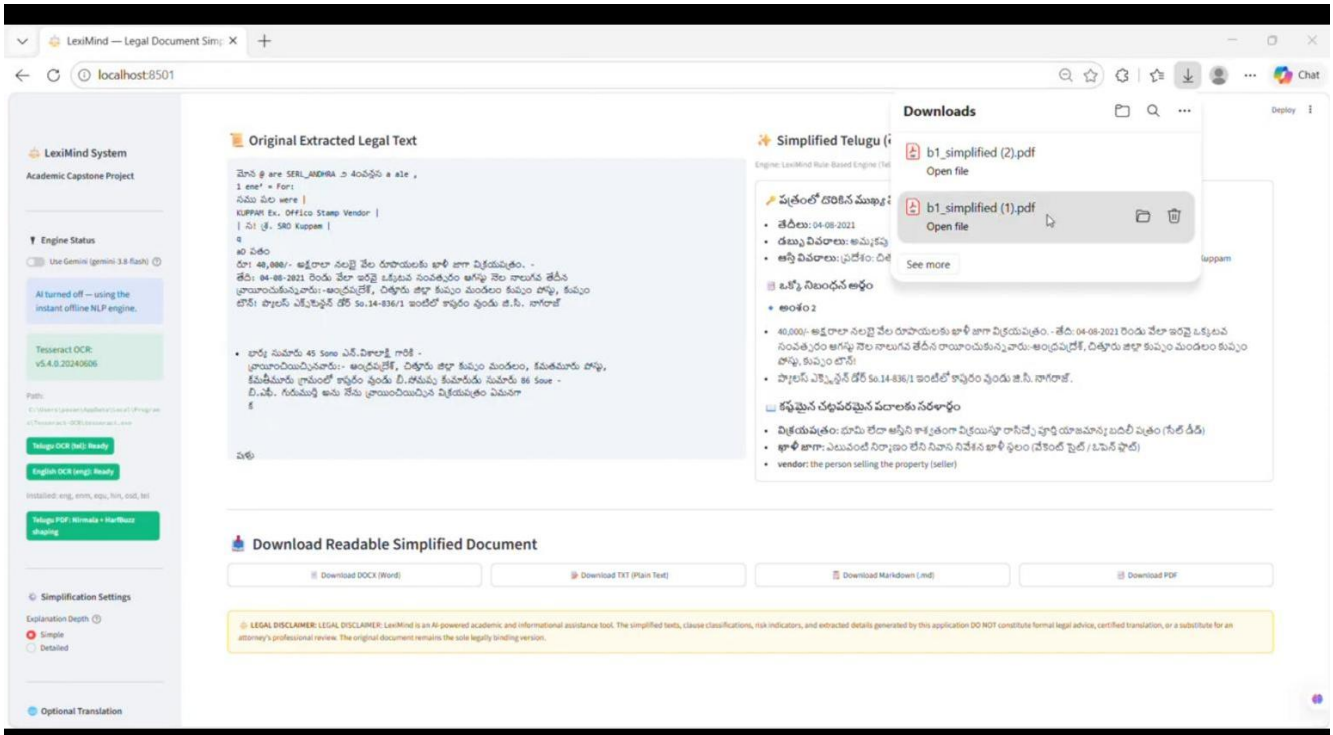


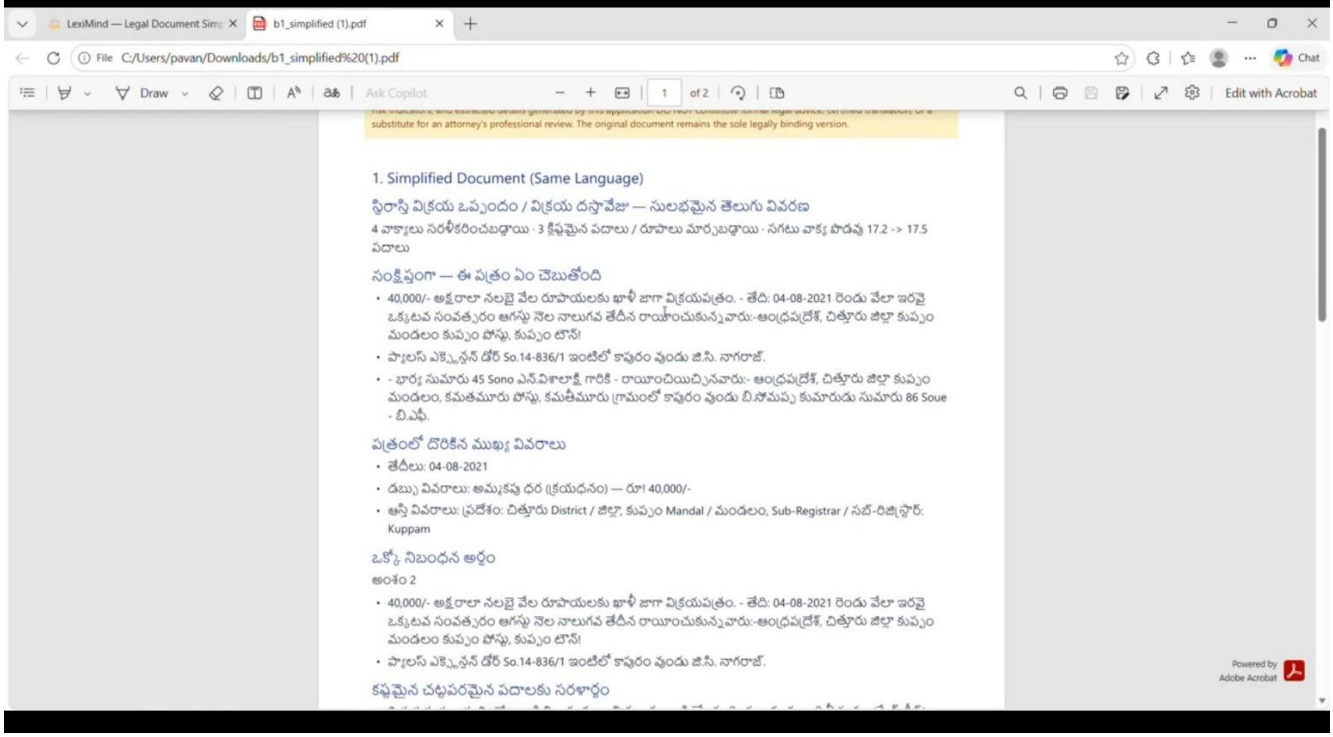

LexiMind — Legal Document Simp
localhost:8501
Chat
Deploy
LexiMind System
Academic Capstone Project
Engine Status
Use Gemini (gemini-3.8-flash)
AI turned off — using the instant offline NLP engine.
Tesseract OCR: v5.4.0.20240606
Telugu OCR (tel): Ready
English OCR (eng): Ready
Installed: eng, enm, equ, hin, osd, tel
Telugu PDF: Nirmala + HarfBuzz shaping
Simplification Settings
Explanation Depth
Simple
Detailed
Optional Translation
Extracted b1.jpeg via image_ocr (tel+eng) (110 words)
Step 1: Inspect, Correct & Prepare Legal Text
Review and fix any OCR or transcription errors below, or paste legal text directly:
Original Extracted Text (Editable for OCR error correction or manual input)
Detected Language
Telugu (తెలుగు)
Telugu Character %
85.8%
English Character %
14.2%
Word Count
110
Step 2: Generate Same-Language Simplification
Primary Action: Simplifies from Telugu (తెలుగు) into Simplified Telugu (తెలుగు) (Same-Language Simplification).
Simplify Legal Document Now
LEGAL DISCLAIMER: LEGAL DISCLAIMER: LexiMind is an AI-powered academic and informational assistance tool. The simplified texts, clause classifications, risk indicators, and extracted details generated by this application DO NOT constitute formal legal advice, certified translation, or a substitute for an attorney's professional review. The original document remains the sole legally binding version.

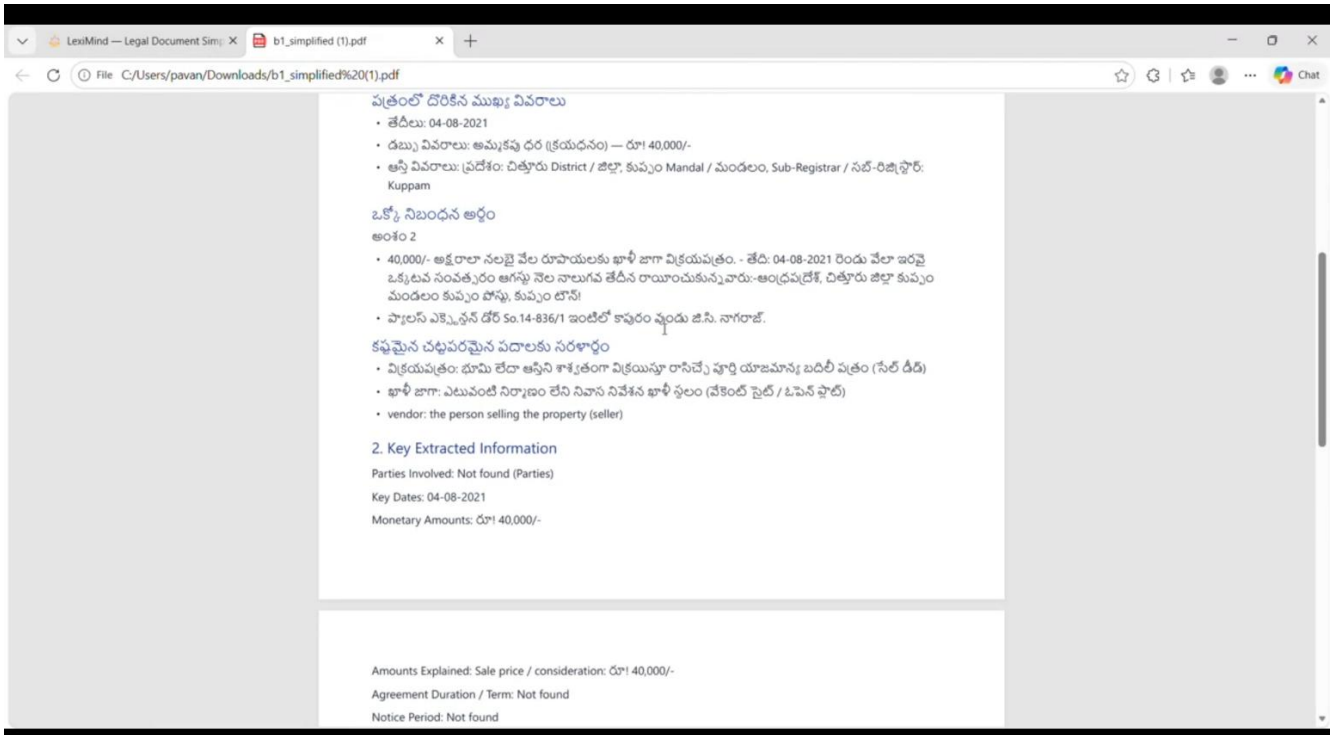

LexiMind — Legal Document Simp
localhost:8501
Chat
Stop
Deploy
LexiMind System
Academic Capstone Project
Engine Status
Use Gemini (gemini-3.8-flash)
AI turned off — using the instant offline NLP engine.
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Telugu OCR (tel): Ready
English OCR (eng): Ready
Installed: eng, enm, equ, hin, osd, tel
Telugu PDF: Nirmala + HarfBuzz shaping
Simplification Settings
Explanation Depth
Simple
Step 2: Generate Same-Language Simplification
Primary Action: Simplifies from Telugu (తెలుగు) into Simplified Telugu (తెలుగు) (Same-Language Simplification).
Simplify Legal Document Now
Simplification Results: Original vs. Simplified Comparison
Sentences Simplified
4
Legal Phrases Rewritten
3
Key Facts Found
2
Detected Language
Telugu (తెలుగు)
Original Extracted Legal Text
Simplified Telugu (తెలుగు) (Simple)
Engine: LexiMind Rule-Based Engine (Telugu Legal NLP Heuristics — Offline Mode)
సంక్షిప్తంగా – ఈ పత్రం ఏం చెబుతోంది
పత్రంలో దొరికిన ముఖ్య వివరాలు
ఒక్క నిబంధన అర్థం


LexiMind — Legal Document Simp
localhost:8501
Chat
Deploy
LexiMind System
Academic Capstone Project
Engine Status
Use Gemini (gemini-3.0-flash)
AI turned off — using the instant offline NLP engine.
Tesseract OCR: v5.4.0.20240606
Telugu OCR (tel): Ready
English OCR (eng): Ready
Telugu PDF: Nirmala + HarfBuzz shaping
Simplification Settings
Simplification Results: Original vs. Simplified Comparison
Sentences Simplified
4
Legal Phrases Rewritten
3
Key Facts Found
2
Detected Language
Telugu (తెలుగు)
Original Extracted Legal Text
Simplified Telugu (తెలుగు) (Simple)
Engine: LexiMind Rule-Based Engine (Telugu Legal NLP Heuristics — Offline Mode)
పత్రంలో దొరికిన ముఖ్య వివరాలు
తేదీలు: 04-08-2021
డబ్బు వివరాలు: అమ్మకపు ధర (క్రయధనం) — రూ! 40,000/-
ఆస్తి వివరాలు: ప్రదేశం: చిత్తూరు District / జిల్లా, కుప్పం Mandal / మండలం, Sub-Registrar / సబ్-రిజిస్ట్రార్: Kuppam
ఒక్కో నిబంధన అర్థం
అంశం 2
40,000/- అక్షరాలా నలభై వేల రూపాయలకు భాళీ జాగా విక్రయపత్రం. - తేదీ: 04-08-2021 రెండు వేలా ఇరవై ఒక్కటవ సంవత్సరం ఆగస్టు నెల నాలుగవ తేదీన రాయించుకున్నవారు:-ఆంధ్రప్రదేశ్, చిత్తూరు జిల్లా కుప్పం మండలం కుప్పం టౌన్, కుప్పం టౌన్!
ప్లాట్స్ ఎక్స్టెన్షన్ డోర్ No.14-836/1 ఇంటిలో కాపురం వుండు జి.సీ. నాగరాజ్.
కష్టమైన చట్టపరమైన పదాలకు సరళార్థం
విక్రయపత్రం: భూమి లేదా ఆస్తిని శాశ్వతంగా విక్రయిస్తూ రాసిచ్చే పూర్తి యాజమాన్య బదిలీ పత్రం (సేల్ డీడ్)
భాళీ జాగా: ఎటువంటి నిర్మాణం లేని నివాస నివేశన భాళీ స్థలం (వేకెంట్ ప్లాట్ / ఓపెన్ ప్లాట్)
vendor: the person selling the property (seller)


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Engine Status
Use Gemini (gemini-3.0-flash)
AI turned off — using the instant offline NLP engine.
Tesseract OCR: v5.4.0.20240606
Telugu OCR (tel): Ready
English OCR (eng): Ready
Original Extracted Legal Text
Simplified Telugu (తెలుగు) (Simple)
Engine: LexiMind Rule-Based Engine (Telugu Legal NLP Heuristics — Offline Mode)
పత్రంలో దొరికిన ముఖ్య వివరాలు
తేదీలు: 04-08-2021
డబ్బు వివరాలు: అమ్మకపు ధర (క్రయధనం) — రూ! 40,000/-
ఆస్తి వివరాలు: ప్రదేశం: చిత్తూరు District / జిల్లా, కుప్పం Mandal / మండలం, Sub-Registrar / సబ్-రిజిస్ట్రార్: Kuppam
ఒక్కో నిబంధన అర్థం
అంశం 2
40,000/- అక్షరాలా నలభై వేల రూపాయలకు భాళీ జాగా విక్రయపత్రం. - తేదీ: 04-08-2021 రెండు వేలా ఇరవై ఒక్కటవ సంవత్సరం ఆగస్టు నెల నాలుగవ తేదీన రాయించుకున్నవారు:-ఆంధ్రప్రదేశ్, చిత్తూరు జిల్లా కుప్పం మండలం కుప్పం టౌన్, కుప్పం టౌన్!
ప్లాట్స్ ఎక్స్టెన్షన్ డోర్ No.14-836/1 ఇంటిలో కాపురం వుండు జి.సీ. నాగరాజ్.
కష్టమైన చట్టపరమైన పదాలకు సరళార్థం
విక్రయపత్రం: భూమి లేదా ఆస్తిని శాశ్వతంగా విక్రయిస్తూ రాసిచ్చే పూర్తి యాజమాన్య బదిలీ పత్రం (సేల్ డీడ్)
భాళీ జాగా: ఎటువంటి నిర్మాణం లేని నివాస నివేశన భాళీ స్థలం (వేకెంట్ ప్లాట్ / ఓపెన్ ప్లాట్)
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AI turned off -- using the instant offline NLP engine.
Tesseract OCR: v5.4.0.20240606
Telugu OCR (tel): Ready
English OCR (eng): Ready
Telugu PDF: Nirmala + HarfBuzz shaping
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1. Simplified Document (Same Language)
స్థిరాస్తి విక్రయ ఒప్పందం / విక్రయ దస్తావేజు — సులభమైన తెలుగు వివరణ
4 వాక్యాలు సరళీకరించబడ్డాయి · 3 క్లిష్టమైన పదాలు / రూపాలు మార్చబడ్డాయి · సగటు వాక్య పొడవు 17.2 -> 17.5 పదాలు
సంక్షిప్తంగా — ఈ పత్రం ఏం చెబుతోంది
40,000/- అక్షరాలా నలబై వేల రూపాయలకు భాళీ జాగా విక్రయపత్రం. - తేదీ: 04-08-2021 రెండు వేలా ఇరవై ఒక్కటవ సంవత్సరం ఆగస్టు నెల నాలుగవ తేదీన రాయించుకున్నవారు:-ఆంధ్రప్రదేశ్, చిత్తూరు జిల్లా కుప్పం మండలం కుప్పం పోస్టు, కుప్పం టౌన్!
ప్లాట్ నెంబర్ 45 Sono ఎన్.విశాలాక్షి గారికి - రాయించియిచ్చినవారు:- ఆంధ్రప్రదేశ్, చిత్తూరు జిల్లా కుప్పం మండలం, కమతమూరు పోస్టు, కమతీమూరు గ్రామంలో కాపురం వుండు బి.సోమప్ప కుమారుడు సుమారు 86 Soue - బి.ఎఫ్.
పత్రంలో దొరికిన ముఖ్య వివరాలు
తేదీలు: 04-08-2021
డబ్బు వివరాలు: అమ్మకపు ధర (క్రయధనం) — రూ! 40,000/-
ఆస్తి వివరాలు: ప్రదేశం: చిత్తూరు District / జిల్లా, కుప్పం Mandal / మండలం, Sub-Registrar / సబ్-రిజిస్ట్రార్: Kuppam
ఒక్కో నిబంధన అర్థం
అంశం 2
40,000/- అక్షరాలా నలబై వేల రూపాయలకు భాళీ జాగా విక్రయపత్రం. - తేదీ: 04-08-2021 రెండు వేలా ఇరవై ఒక్కటవ సంవత్సరం ఆగస్టు నెల నాలుగవ తేదీన రాయించుకున్నవారు:-ఆంధ్రప్రదేశ్, చిత్తూరు జిల్లా కుప్పం మండలం కుప్పం పోస్టు, కుప్పం టౌన్!
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కష్టమైన చట్టపరమైన పదాలకు సరళార్థం
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పత్రంలో దొరికిన ముఖ్య వివరాలు

- తేదీలు: 04-08-2021
- డబ్బు వివరాలు: అమ్మకపు ధర (క్రయధనం) — రూ! 40,000/-
- ఆస్తి వివరాలు: ప్రదేశం: చిత్తూరు District / జిల్లా, కుప్పం Mandal / మండలం, Sub-Registrar / సబ్-రిజిస్ట్రార్: Kuppam

ఒక్కో నిబంధన అర్థం

అంశం 2

- 40,000/- అక్షరాలా నలభై వేల రూపాయలకు ఖాళీ జాగా విక్రయపత్రం. - తేదీ: 04-08-2021 రెండు వేలా ఇరవై ఒక్కటవ సంవత్సరం ఆగస్టు నెల నాలుగవ తేదీన రాయించుకున్నవారు:-ఆంధ్రప్రదేశ్, చిత్తూరు జిల్లా కుప్పం మండలం కుప్పం పోస్టు, కుప్పం టౌన్!
- ప్లాట్ నెం.14-836/1 ఇంటిలో కాపురం వుండు జి.సి. నాగరాజ్.

కష్టమైన చట్టపరమైన పదాలకు సరళార్థం

- విక్రయపత్రం: భూమి లేదా ఆస్తిని శాశ్వతంగా విక్రయిస్తూ రాసిచ్చే పూర్తి యాజమాన్య బదిలీ పత్రం (సేల్ డీడ్)
- ఖాళీ జాగా: ఎటువంటి నిర్మాణం లేని నివాస నివేశన ఖాళీ స్థలం (వేకెంట్ సైట్ / ఓపెన్ ప్లాట్)
- vendor: the person selling the property (seller)

2. Key Extracted Information

Parties Involved: Not found (Parties)

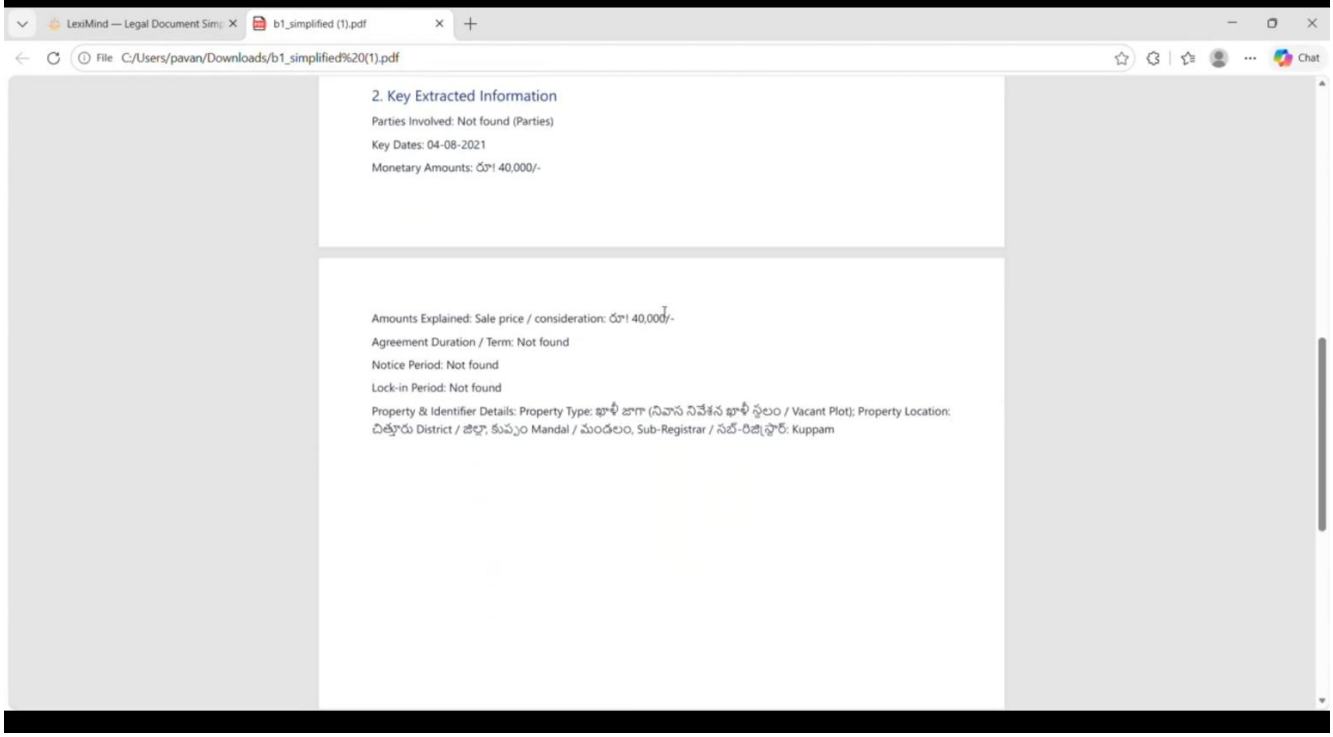
Key Dates: 04-08-2021

Monetary Amounts: రూ! 40,000/-

Amounts Explained: Sale price / consideration: రూ! 40,000/-

Agreement Duration / Term: Not found

Notice Period: Not found


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2. Key Extracted Information

Parties Involved: Not found (Parties)

Key Dates: 04-08-2021

Monetary Amounts: రూ! 40,000/-

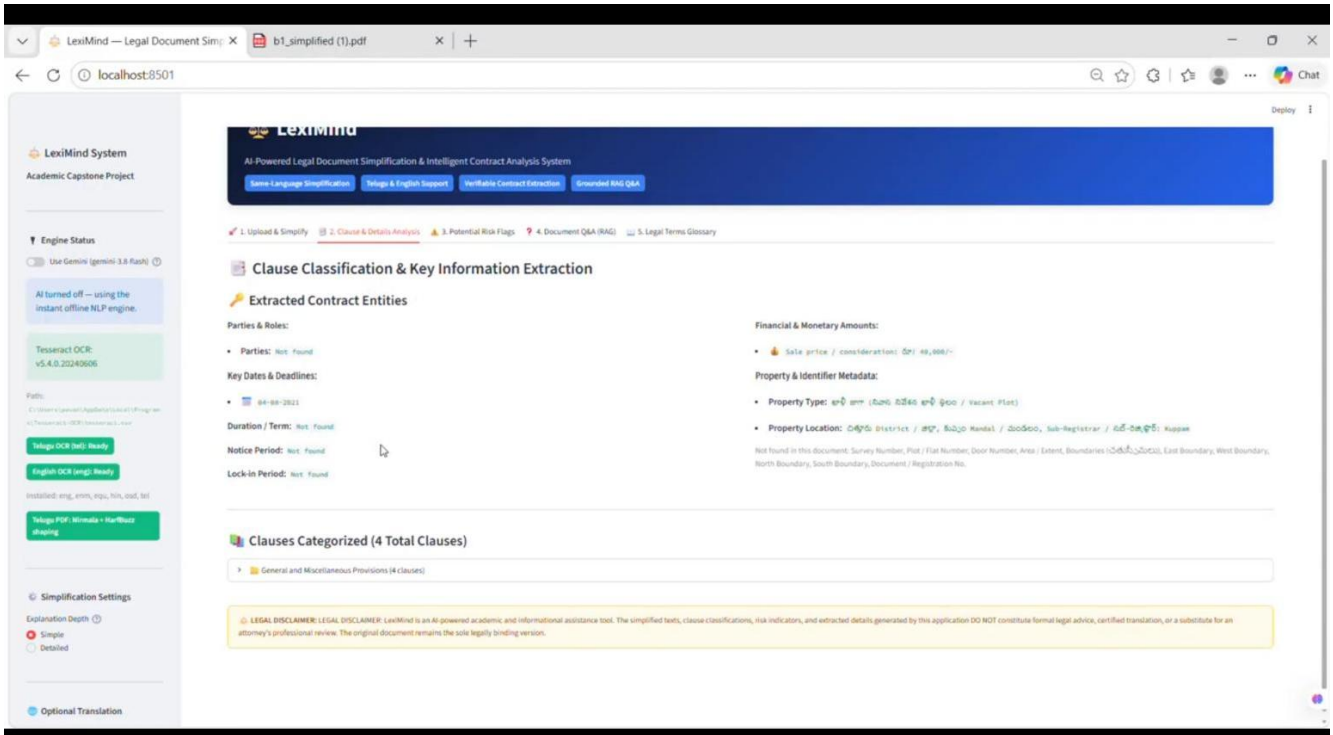
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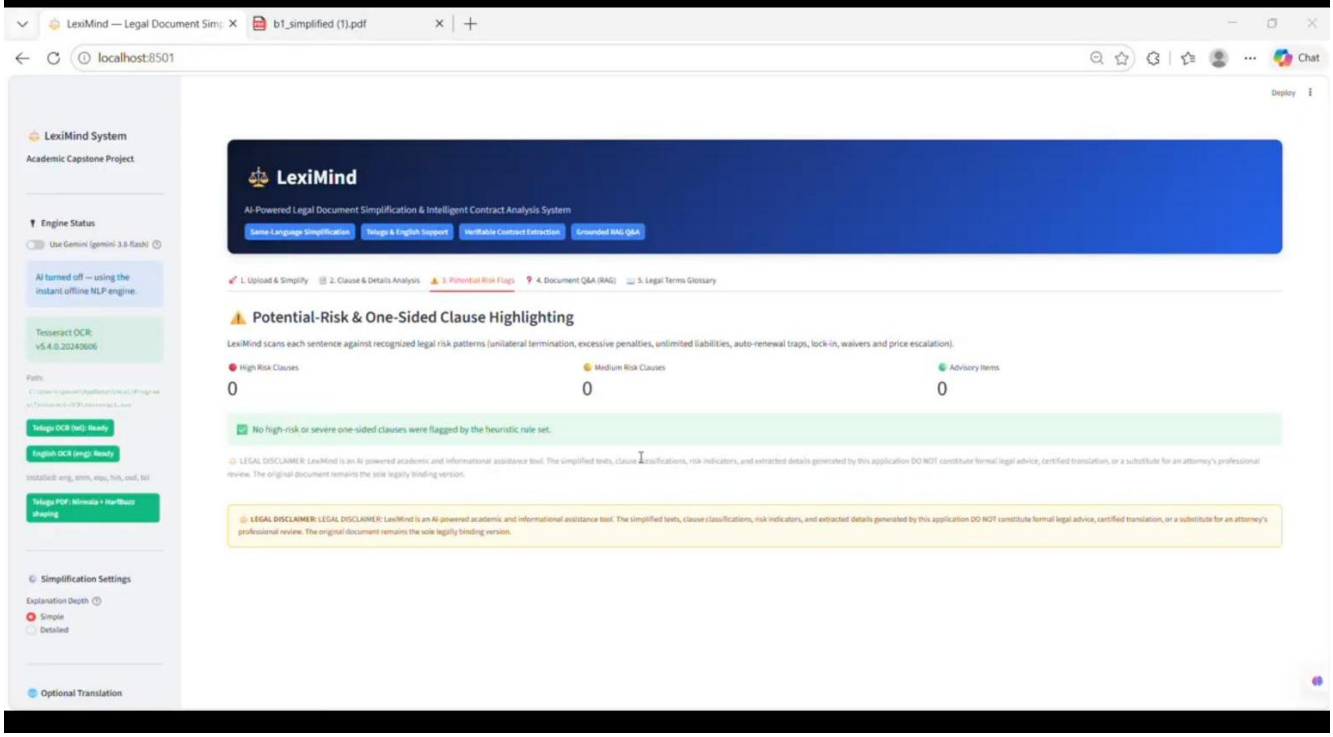
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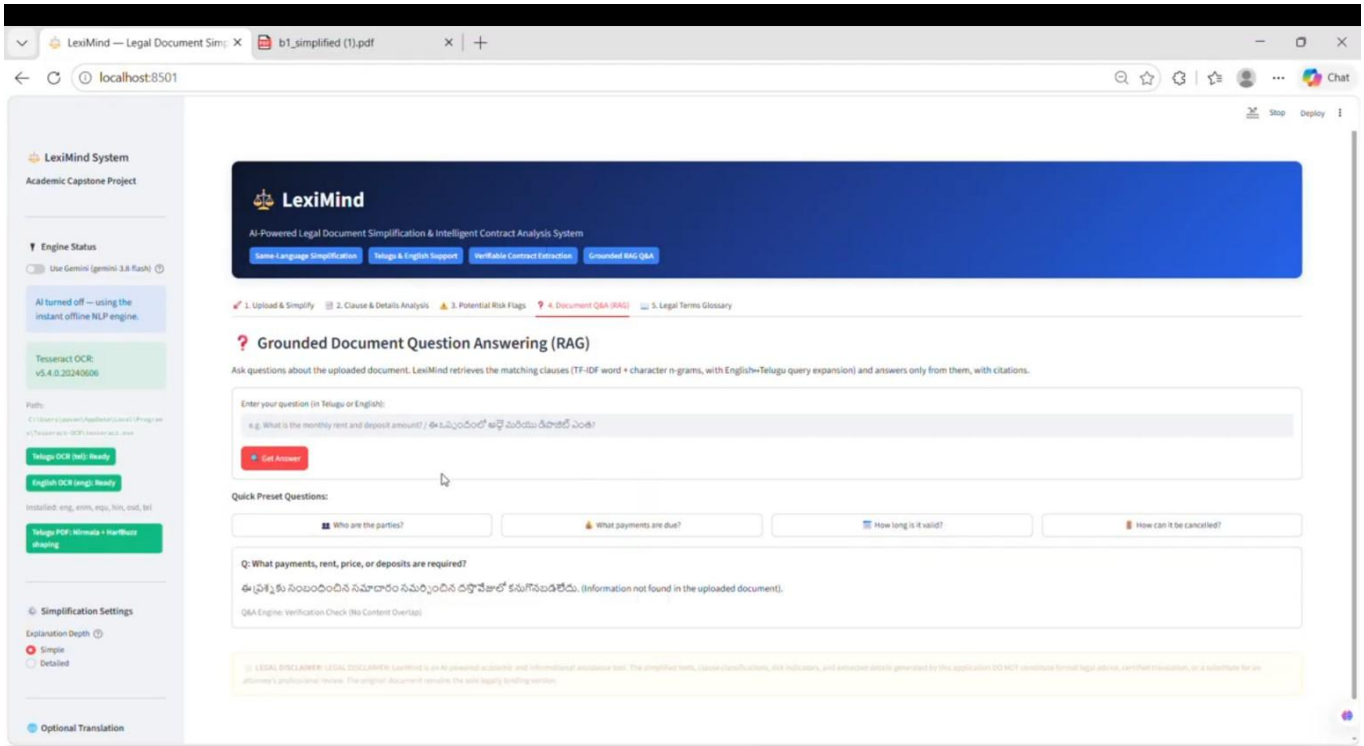
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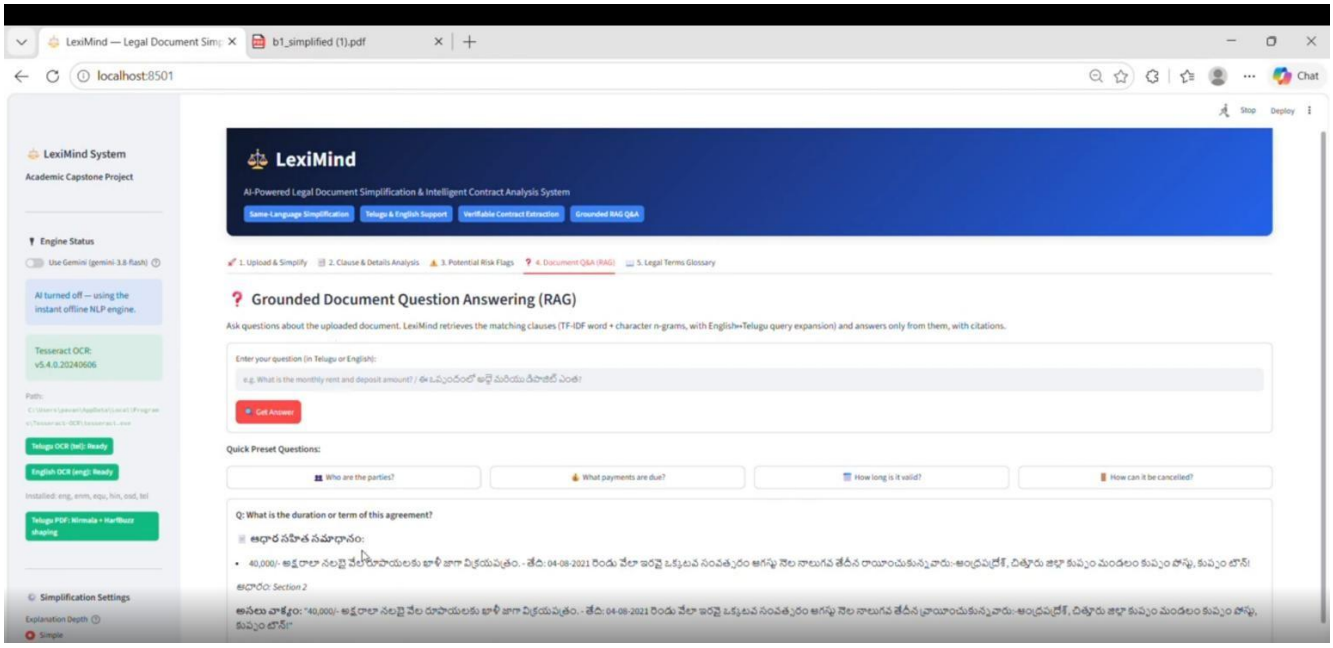
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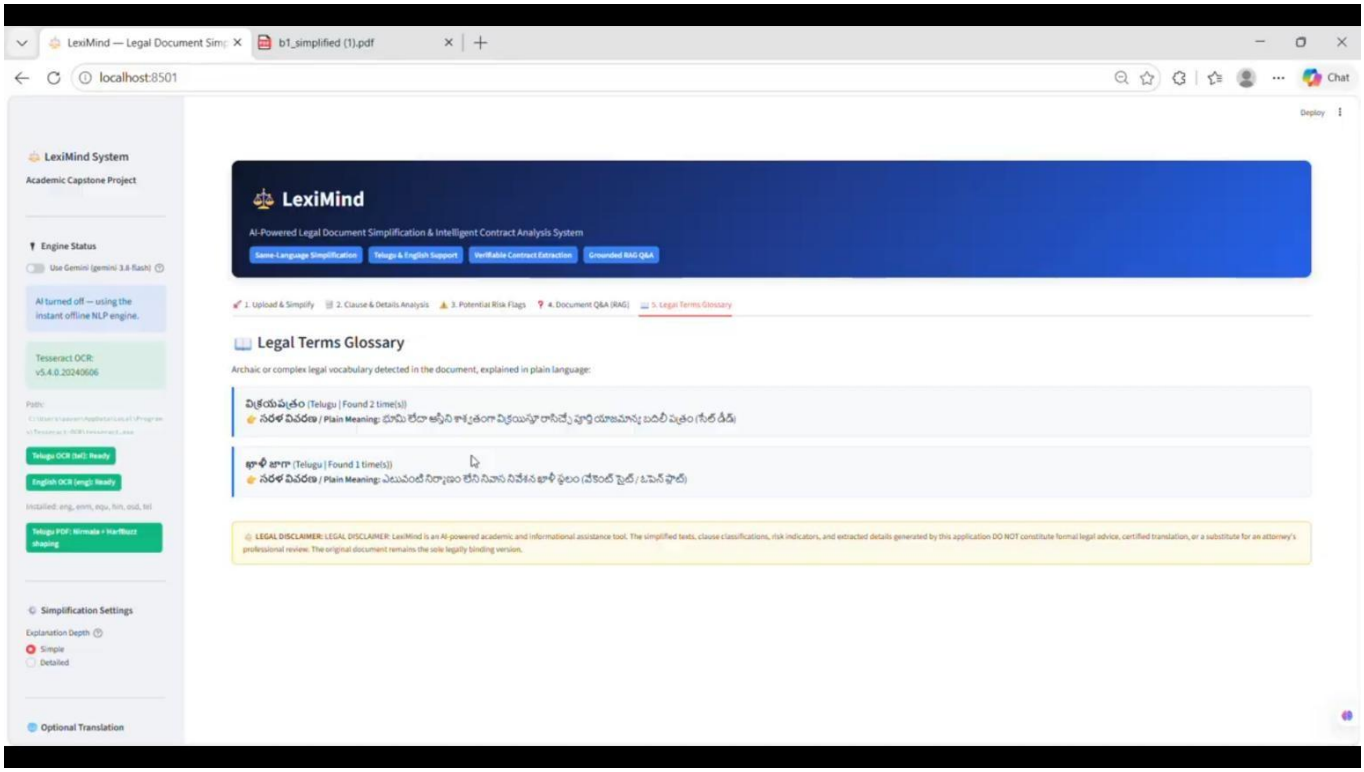
Property & Identifier Details: Property Type: ఖాళీ జాగా (నివాస నివేశన ఖాళీ స్థలం / Vacant Plot); Property Location: చిత్తూరు District / జిల్లా, కుప్పం Mandal / మండలం, Sub-Registrar / సబ్-రిజిస్ట్రార్: Kuppam


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Grounded RAG Q&A
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2. Clause & Details Analysis
3. Potential Risk Flags
4. Document Q&A (RAG)
5. Legal Terms Glossary
Clause Classification & Key Information Extraction
Extracted Contract Entities
Parties & Roles:
Parties: Not found
Key Dates & Deadlines:
04-08-2021
Duration / Term: Not found
Notice Period: Not found
Lock-in Period: Not found
Financial & Monetary Amounts:
Sale price / consideration: ₹₹! 40,000/-
Property & Identifier Metadata:
Property Type: ఖాళీ జాగా (మిగతా వివరాలు ఖాళీ స్థలం / Vacant Plot)
Property Location: చిత్తూరు District / జిల్లా, పుంగనూరు Mandal / మండలం, Sub-Registrar / సబ్-రిజిస్ట్రార్: Kuppam
Not found in this document: Survey Number, Plot / Flat Number, Door Number, Area / Extent, Boundaries (నాలుగు వైపులు), East Boundary, West Boundary, North Boundary, South Boundary, Document / Registration No.
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General and Miscellaneous Provisions (4 clauses)
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4. Document Q&A (RAG)
5. Legal Terms Glossary
Potential-Risk & One-Sided Clause Highlighting
LexiMind scans each sentence against recognized legal risk patterns (unilateral termination, excessive penalties, unlimited liabilities, auto-renewal traps, lock-in, waivers and price escalation).
High Risk Clauses
0
Medium Risk Clauses
0
Advisory Items
0
No high-risk or severe one-sided clauses were flagged by the heuristic rule set.
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Ask questions about the uploaded document. LexiMind retrieves the matching clauses (TF-IDF word + character n-grams, with English↔Telugu query expansion) and answers only from them, with citations.
Enter your question (in Telugu or English):
e.g. What is the monthly rent and deposit amount? / ఈ ఒప్పందంలో అద్దె మరియు డిపాజిట్ ఎంత?
Get Answer
Quick Preset Questions:
Who are the parties?
What payments are due?
How long is it valid?
How can it be cancelled?
Q: What payments, rent, price, or deposits are required?
ఈ ప్రశ్నకు సంబంధించిన సమాచారం సమర్పించిన దస్తావేజులో కనుగొనబడలేదు. (Information not found in the uploaded document).
Q&A Engine: Verification Check (No Content Overlap)


LexiMind — Legal Document Simp
b1_simplified (1).pdf
localhost:8501
Chat
Stop
Deploy
LexiMind System
Academic Capstone Project
Engine Status
Use Gemini (gemini-3.8-flash)
AI turned off — using the instant offline NLP engine.
Tesseract OCR: v5.4.0.20240606
Telugu OCR (tel): Ready
English OCR (eng): Ready
Telugu PDF: Nirmala + HarfBuzz shaping
Simplification Settings
Explanation Depth
Simple
LexiMind
AI-Powered Legal Document Simplification & Intelligent Contract Analysis System
Same-Language Simplification
Telugu & English Support
Verifiable Contract Extraction
Grounded RAG Q&A
1. Upload & Simplify
2. Clause & Details Analysis
3. Potential Risk Flags
4. Document Q&A (RAG)
5. Legal Terms Glossary
Grounded Document Question Answering (RAG)
Ask questions about the uploaded document. LexiMind retrieves the matching clauses (TF-IDF word + character n-grams, with English↔Telugu query expansion) and answers only from them, with citations.
Enter your question (in Telugu or English):
e.g. What is the monthly rent and deposit amount? / ఈ ఒప్పందంలో అద్దె మరియు డిపాజిట్ ఎంత?
Get Answer
Quick Preset Questions:
Who are the parties?
What payments are due?
How long is it valid?
How can it be cancelled?
Q: What is the duration or term of this agreement?
ఆధార సహిత సమాధానం:
40,000/- అక్షరాలా నలభై వేల రూపాయలకు భాళీ జాగా విక్రయపత్రం. - తేదీ: 04-08-2021 రిందు వేలా ఇరవై ఒక్కటవ సంవత్సరం ఆగస్టు నెల నాలుగవ తేదీన రాయించుకున్నవారు:-ఆంధ్రప్రదేశ్, చిత్తూరు జిల్లా కుప్పం మండలం కుప్పం పాస్టు, కుప్పం టౌన్!
ఆధారం: Section 2
అసలు వాక్యం: "40,000/- అక్షరాలా నలభై వేల రూపాయలకు భాళీ జాగా విక్రయపత్రం. - తేదీ: 04-08-2021 రిందు వేలా ఇరవై ఒక్కటవ సంవత్సరం ఆగస్టు నెల నాలుగవ తేదీన వ్రాయించుకున్నవారు:-ఆంధ్రప్రదేశ్, చిత్తూరు జిల్లా కుప్పం మండలం కుప్పం పాస్టు, కుప్పం టౌన్!"



10. REFERENCES

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11. OpenAI. OpenAI API Documentation. https://platform.openai.com/docs/
12. PyMuPDF. PyMuPDF Documentation. https://pymupdf.readthedocs.io/
13. Python Software Foundation. Python Documentation. https://docs.python.org/3/

PROJECT PPT LINK:

https://docs.google.com/presentation/d/1AzX_G6caGcXLZgsuB4T-rwzmQ9ULucdk/edit?usp=sharing&ouid=100961027915246184268&rtpof=true&sd=true

PROJECT DEMO VIDEO LINK:

https://drive.google.com/file/d/1b_eAN9DQm31SzMZlzXwzFSQxkOmYeNC3/view?usp=drive_link

PROJECT CLASS PRESENTATION:



